

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide $K_4Fe(CN)_6$. After filtering, the precipitate is dried and weighed. The reactions which occur are $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$ $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$ If a 2 gram sample of rock yields 0.969 gram of $Zn_2Fe(CN)_6$, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight $Zn_2Fe(CN)_6 = 342.6$.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<reasoning process here >

We are given :

- A 2 gram sample of rock yields 0.969 grams of $Zn_2Fe(CN)_6$.
- We are to find the percentage of Zn in the original rock sample .
- The relevant reactions are :
1 . $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$
2 . $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$

From the stoichiometry :

- 2 moles of $ZnCl_2$ produce 1 mole of $Zn_2Fe(CN)_6$.
- Each mole of $Zn_2Fe(CN)_6$ contains 2 moles of Zn .

So , we can work backwards from the mass of $Zn_2Fe(CN)_6$ to find the mass of Zn in it , and then find the percentage of Zn in the original rock sample .

Given :

- Molecular weight of $Zn_2Fe(CN)_6 = 342.6$ g/mol
- Atomic weight of Zn = 65.4 g/mol
- So , mass of Zn in 1 mole of $Zn_2Fe(CN)_6 = 2 \times 65.4 = 130.8$ g

So , fraction of Zn in $Zn_2Fe(CN)_6 = 130.8 / 342.6$

Now , mass of Zn in 0.969 g of $Zn_2Fe(CN)_6$:
 $= 0.969 \times (130.8 / 342.6)$

Calculate that :

First , $130.8 / 342.6 \approx 0.3818$

Then :
 $0.969 \times 0.3818 \approx ?$

$0.969 \times 0.3818 \approx 0.370$

So , mass of Zn in the sample ≈ 0.370 g

Original sample mass = 2 g

So , % Zn = $(0.370 / 2) \times 100 = 18.5\%$

That is approximately 18.5% , which is closest to :

F. 18% Zn

But let's do the calculation more precisely .

Step-by-step :

- 1 . Mass of Zn in $Zn_2Fe(CN)_6$:
- Molar mass $Zn_2Fe(CN)_6 = 342.6$
- Mass of Zn = $2 \times 65.4 = 130.8$
- Fraction of Zn = $130.8 / 342.6 = ?$

$130.8 \div 342.6 = ?$

$342.6 \times 0.3818 = 342.6 \times 0.38 = 130.188$, and $342.6 \times 0.0018 = \sim 0.616$, total ~ 130.804 $\approx$ very close .

So , $130.8 / 342.6$